Writing equations

Aim

To write word equations for reactions.

Equipment

- Resource sheet E6a
- scissors

What to do

1. Cut out the cards on the resource sheet.
2. Arrange the cards for hydrogen and chloride side by side:

H **hydrogen**	**Cl** **chloride**

This represents hydrochloric acid. All acids have hydrogen as part of the formula. Sulphuric acid is hydrogen sulphate and nitric acid is hydrogen nitrate.

3. Now arrange the cards for sodium and hydroxide side by side:

Na **sodium**	**OH** **hydroxide**

This represents the alkali sodium hydroxide. All alkalis have hydroxide as part of the formula.

4. You are going to write the word equation for the neutralisation reaction between hydrochloric acid and sodium hydroxide.

   First write the reactants on the left of the equation:

   hydrochloric acid + **sodium hydroxide** →

5. Now rearrange the cards to show the products:

Na **sodium**	**Cl** **chloride**

H **hydrogen**	**OH** **hydroxide**

6. Now complete the word equation:

   hydrochloric acid + **sodium hydroxide** → **sodium chloride** + **water**

   Note that hydrogen hydroxide is water, H_2O.

Continued ⟶

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Writing equations continued

Questions

Use the same method to write a word equation for each of the following reactions.

1 Nitric acid + potassium hydroxide

2 Magnesium + hydrochloric acid

3 Copper oxide + sulphuric acid

4 Zinc carbonate + hydrochloric acid

5 Zinc + sulphuric acid

6 Nitric acid + magnesium hydroxide

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NO_3 nitrate	C carbon
NO_3 nitrate	SO_4 sulphate
Na sodium	CO_3 carbonate
Na sodium	K potassium
K potassium	H hydrogen
Cu copper	H hydrogen
Mg magnesium	O oxide
Zn zinc	Cl chloride
OH hydroxide	
OH hydroxide	
Cl chloride	

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